

CS23532-COMPUTER NETWORKS

EXPERIMENT -12

End –End Communication at Transport Layer

- a) Implement echo client server using TCP/UDP sockets.
- b) Implement a chat program using socket programming.

```
tcp_server.py import socket server =
socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("127.0.0.1", 5000)) server.listen(1) print("TCP
Echo Server started...") conn, addr = server.accept()
print("Connected:", addr)
while True:
data = conn.recv(1024) if not
data: break
print("Client:", data.decode())
conn.send(data)
conn.close() server.close()

tcp_client.py import
socket
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("127.0.0.1", 5000))
print("Connected to TCP Echo Server")
while True:
msg = input("Enter message: ")
if msg.lower() == "exit":
break
client.send(msg.encode())
```

```
data = client.recv(1024)
```

```
print("Server:", data.decode())  
client.close()
```

```
udp_server.py import socket  
  
server = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)  
  
server.bind(("127.0.0.1", 5001))  
  
print("UDP Echo Server started...")  
  
while True:  
    data, addr = server.recvfrom(1024)  
  
    print("Client:", data.decode())  
  
    if data.decode().lower() == "exit":  
        break  
  
    server.sendto(data, addr)  
  
server.close()
```

```
udp_client.py import  
socket  
  
client = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)  
  
print("Connected to UDP Echo Server")  
  
while True:  
    msg = input("Enter message: ")  
  
    client.sendto(msg.encode(), ("127.0.0.1", 5001))  
  
    if msg.lower() == "exit":  
        break  
  
    data, addr = client.recvfrom(1024)  
  
    print("Server:", data.decode())  
  
client.close()
```

```

chat_server.py import socket server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("127.0.0.1", 5002)) server.listen(1)

print("Chat Server started...") conn, addr = server.accept()

print("Connected:", addr) while True:

    msg = conn.recv(1024).decode()

    if msg.lower() == "exit":

        break

    print("Client:", msg)

    reply = input("You: ")

conn.send(reply.encode())

    if reply.lower() == "exit":

        break

conn.close() server.close()

chat_client.py import

socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

client.connect(("127.0.0.1", 5002))

print("Connected to Chat Server")

while True:

    msg = input("You: ")

client.send(msg.encode())

    if msg.lower() == "exit":

        break

    reply = client.recv(1024).decode()

print("Server:", reply)

    if reply.lower() == "exit":

        break client.close()

```


The image shows a VS Code editor window with a file explorer on the left and a terminal at the bottom. The file explorer shows a project named 'CN-12' with files: chat_client.py, chat_server.py, tcp_client.py, tcp_server.py, udp_client.py, and udp_server.py. The editor displays the code for chat_client.py, which is a simple socket-based chat client. The terminal shows the execution of three Python scripts: tcp_server.py, tcp_client.py, and udp_client.py. The output of these scripts shows a successful connection to a TCP Echo Server and a UDP Echo Server, with messages being sent and received.

```
1 import socket
2
3 client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
4 client.connect(("127.0.0.1", 5002))
5
6 print("Connected to chat Server")
7
8 while True:
9     msg = input("You: ")
10    client.send(msg.encode())
11
12    if msg.lower() == "exit":
13        break
14
15    reply = client.recv(1024).decode()
16    print("Server:", reply)
17
18    if reply.lower() == "exit":
19        break
20
```

Terminal Output:

```
PS C:\Users\Seeralan M\OneDrive\Documents\CN-12> python tcp_server.py
TCP Echo Server started...
Connected: ("127.0.0.1", 5002)
client: CN LAB Seeralan M
Enter message: Seeralan M
Server: CN LAB Seeralan M
Enter message:

PS C:\Users\Seeralan M\OneDrive\Documents\CN-12> python tcp_client.py
Connected to TCP Echo Server
Enter message: Seeralan M
Server: CN LAB Seeralan M
Enter message:

PS C:\Users\Seeralan M\OneDrive\Documents\CN-12> python udp_server.py
UDP Echo Server started...
Client: CN LAB EXP
Enter message:

PS C:\Users\Seeralan M\OneDrive\Documents\CN-12> python udp_client.py
Connected to UDP Echo Server
Enter message: CN LAB EXP
Server: CN LAB EXP
Enter message:
```

```
PS C:\Users\Seeralan M\OneDrive\Documents\CN-12> python chat_server.py
Chat Server started...
Connected: ('127.0.0.1', 58511)
Client: HII
You: GOOD MORNING
Client: WELCOME
You: 
```

Problems Output **Terminal** Ports Debug Console

```
PS C:\Users\Seeralan M\OneDrive\Documents\CN-12> python chat_client.py
Connected to chat Server
You: HII
Server: GOOD MORNING
You: WELCOME

```